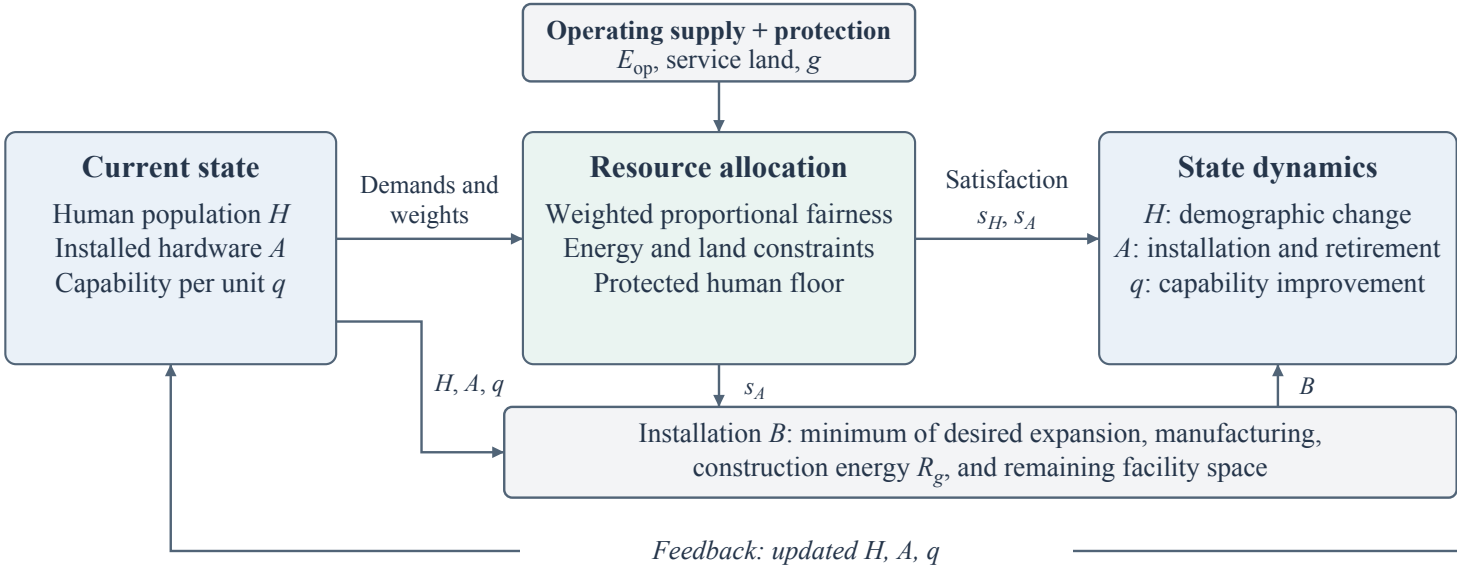

Operating supply + protection
E_{op}, service land, g
Current state
Human population H
Installed hardware A
Capability per unit q
Demands and weights
Resource allocation
Weighted proportional fairness
Energy and land constraints
Protected human floor
Satisfaction
s_H, s_A
State dynamics
H: demographic change
A: installation and retirement
q: capability improvement
H, A, q
s_A
B
Installation B: minimum of desired expansion, manufacturing, construction energy R_g, and remaining facility space
Feedback: updated H, A, q